

Mohammad Tanvir Hasan

✉ Email | ☎ (+1) 501-716-7449 | [in LinkedIn](#) |
[Github](#) | [Website](#)

EDUCATION

M.Sc. in Computer Science
University of Arkansas at Little Rock

Aug 24 - Present
Arkansas, USA

GPA: 4.00/4.00

B.Sc. in Computer Science and Engineering
Khulna University of Engineering & Technology

Grad. 2023
Khulna, Bangladesh

GPA: 3.32/4.00

WORK EXPERIENCE

University of Arkansas at Little Rock
Graduate Assistant

Oct 2024 - Present
Arkansas, USA

- Support undergraduate Cybersecurity courses, develop curriculum, and provide technical guidance to 30+ students.
- Create hands-on lab assignments for Digital Forensics using Wireshark, Autopsy, and EnCase.

Turfhub
Software Engineer

Dec 2022 - Jul 2024
turfhubbd.com

- Built cross-platform app using React Native and NextJS, serving 40+ sports venues and 100,000+ users.
- Led frontend implementation of booking, payment, and authentication features, improving user engagement.
- Conducted 50+ code reviews and established quality standards, reducing bugs.
- Optimized performance with lazy loading and code splitting, reducing load time by 30%.

RESEARCH INTERESTS

AI Alignment & Interpretability: Techniques for understanding neural networks, focusing on transparency in language and vision systems.

Safe Code Generation: Enhancing the reliability and security of AI-driven code generation, drawing from both adversarial testing and interpretability insights to ensure produced code is safe, verifiable, and aligned with intended logic.

AI Governance and Responsible Deployment: Investigating frameworks and best practices for deploying advanced AI systems safely in real-world and high-stakes contexts, informed by empirical robustness benchmarks and adversarial evaluations.

Alignment Literature Engagement: Participated in reading groups covering “Shard Theory”, “Eliciting Latent Knowledge (ELK)”, and ARC interpretability papers.

PUBLICATIONS

Conference Proceedings :

- **M. T. Hasan**, K. M. Rokibul Alam, M. A. Habib and M. Navid Razzaque, “Data Trading System Using Blockchain with Oracle Network,” *2023 6th International Conference on Electrical Information and Communication Technology (EICT)*, Khulna, Bangladesh, 2023, pp. 1-6, doi: 10.1109/EICT61409.2023.10427802.
- M. N. Razzaque, K. M. Rokibul Alam, M. A. Habib and **M. Tanvir Hasan**, “Enhancing E-Voting Security with Blockchain-Backed Decentralized Authorization,” *2023 6th International Conference on Electrical Information and Communication Technology (EICT)*, Khulna, Bangladesh, 2023, pp. 1-6, doi: 10.1109/EICT61409.2023.10427881.

RELEVANT COURSES

Artificial Intelligence, Machine Learning, Natural Language Processing, Computer Architecture, Mobile Computing, Digital Logic Design, Digital System Design, Robotics

PROJECTS

Retrieval Augmented Generation | *Langchain, OpenAI, Pinecone*

[Source Code](#)

- Built information retrieval system using OpenAI GPT models with LangChain for pipeline orchestration and Pinecone for vector storage.
- Enhanced QA accuracy by coupling LLMs with external knowledge retrieval.
- [Live Demo](#)

Multi-Agent Recommendation System | *AWS, Anthropic, Python*

[Source Code](#)

- Designed multi-agent system using AWS Bedrock with supervisor agent dynamically invoking collaborator agents.
- Utilized AWS Lambda for request processing and agent orchestration.
- Leveraged S3 buckets for efficient data storage and retrieval.

Vision Transformer vs. EfficientNet B2 Image Classification | *Scikit-learn, Pandas, PyTorch, Gradio*

[Source Code](#)

- Developed a custom Vision Transformer (ViT) from scratch and rigorously compared its performance against a pre-trained EfficientNetB2 classifier on image classification tasks.
- Evaluated model accuracy, efficiency, and inference times, highlighting the trade-offs between self-implemented architectures and state-of-the-art pretrained models.
- [Live Demo](#)

SKILLS

Languages: Python, Javascript, C++, C

Frameworks & Libraries: PyTorch, Hardhat, NextJS

Databases: MySQL, MongoDB, Pinecone

Tools: Docker, MLflow, Jupyter, Figma, Android Studio, Git, LaTeX

Interpersonal: Problem Solving, Mentoring

Language: English, Bengali

EXTRACURRICULAR

Vice President

Oct 24 - Present

Muslim Student Association

- Organized charity events raising over \$2000.